

IDRE FJALL, Sweden

WMO: 023070

Lat: 61.8886N

Lon: 12.8521E

Elev: 2851

StdP: 13.24

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.0	3.3	-6.9	4.2	9.1	-3.1	5.2	7.9	37.2	23.9	33.3	24.3	9.2	130	0.844

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	70.6	56.0	67.2	54.8	63.9	53.5	58.9	66.2	57.1	63.8	55.4	61.1	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.1	74.6	60.2	54.4	69.9	59.1	52.6	65.4	57.7	26.9	66.3	25.7	63.9	24.6	61.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.8	27.6	24.4	DB	-1.8	74.5	6.1	4.2	-6.2	77.5	-9.7	79.9	-13.1	82.3	-17.5	85.3	(4)
(5)			WB	-1.9	60.8	5.6	1.8	-6.0	62.1	-9.3	63.2	-12.4	64.2	-16.5	65.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	19.8	19.4	23.2	32.2	41.4	49.1	54.3	51.5	44.4	34.2	26.1	21.8	(6)	
	DBStd	14.31	7.47	8.50	7.35	6.95	8.00	7.20	6.44	5.70	5.83	6.63	7.13	7.78	(7)	
	HDD50	5889	935	857	831	533	292	100	26	48	185	491	718	873	(8)	
	HDD65	11000	1400	1277	1296	983	733	478	336	418	617	956	1168	1338	(9)	
	CDD50	371	0	0	0	0	24	74	161	95	17	0	0	0	(10)	
	CDD65	9	0	0	0	0	0	2	6	1	0	0	0	0	(11)	
	CDH74	18	0	0	0	0	0	5	13	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	12.1	12.8	12.7	13.4	12.1	12.0	11.7	10.6	10.8	11.8	12.3	12.4	13.2	(14)	
	PrecAvg	27.3	1.8	1.2	1.1	1.5	2.2	3.1	3.4	3.9	2.5	2.4	2.2	1.7	(15)	
(16) Precipitation	PrecMax	36.8	3.7	2.2	2.2	3.7	4.6	6.2	6.0	6.3	4.8	5.9	6.5	2.9	(16)	
	PrecMin	22.0	0.4	0.2	0.1	0.3	0.5	0.2	0.8	1.4	0.2	0.4	0.9	0.2	(17)	
	PrecStd	3.5	0.9	0.5	0.6	0.9	1.1	1.2	1.3	1.3	1.2	1.3	1.2	0.6	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	38.8	41.9	54.2	69.2	73.8	76.0	71.1	62.7	52.6	45.4	38.6	(19)	
		MCWB	31.4	32.4	36.6	43.7	52.0	54.0	59.2	57.3	52.7	49.1	41.1	36.3	(20)	
	2%	DB	34.3	35.3	38.7	49.3	63.2	69.0	71.5	66.7	58.6	47.6	40.9	36.4	(21)	
		MCWB	31.3	32.0	34.2	40.7	49.4	52.9	57.4	55.3	50.6	44.5	38.8	34.5	(22)	
	5%	DB	32.1	32.8	36.1	45.5	58.5	64.5	68.4	63.4	55.3	45.4	37.8	34.2	(23)	
		MCWB	30.4	30.5	32.7	38.7	47.3	51.7	56.7	54.5	49.1	43.1	36.5	32.7	(24)	
	10%	DB	29.9	30.4	33.3	42.2	53.6	60.5	64.8	60.1	52.4	43.1	35.0	32.1	(25)	
		MCWB	28.6	28.6	30.4	36.6	45.0	50.1	54.9	53.4	47.7	41.1	34.1	30.7	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	34.9	37.6	45.6	53.3	58.3	61.3	60.0	55.2	50.5	42.6	37.5	(27)	
		MCDB	35.3	36.8	40.9	50.8	63.2	65.2	70.0	65.5	58.5	52.5	43.7	37.7	(28)	
	2%	WB	32.4	32.6	35.1	41.8	51.1	55.4	59.7	58.0	52.7	45.8	39.1	35.1	(29)	
		MCDB	33.5	34.4	37.9	47.9	61.0	63.2	68.3	62.9	56.6	46.9	40.3	36.1	(30)	
	5%	WB	30.5	30.8	33.0	39.4	48.5	53.2	58.1	56.3	50.7	43.4	36.7	33.2	(31)	
		MCDB	31.6	32.6	35.7	44.7	55.5	62.9	66.2	61.3	53.9	44.7	37.9	34.2	(32)	
	10%	WB	28.6	28.7	30.9	37.3	45.8	51.3	56.4	54.5	48.9	41.4	34.2	30.9	(33)	
		MCDB	29.8	30.3	33.2	41.8	52.8	59.8	63.3	59.2	51.7	42.6	34.8	32.0	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	8.5	9.1	12.2	12.8	12.0	10.6	9.2	6.9	6.1	6.5	(35)	
		MCDBR	8.3	8.8	10.6	12.8	18.0	17.0	15.8	13.8	13.2	9.1	8.9	8.5	(36)	
		MCWBR	7.3	7.6	7.7	7.7	8.9	7.4	7.0	6.6	7.4	7.4	7.9	8.3	(37)	
	5% WB	MCDBR	7.5	8.2	10.2	12.1	16.3	15.1	14.4	11.7	11.7	7.8	8.5	8.3	(38)	
		MCWBR	7.2	7.6	8.0	7.8	9.2	7.5	7.4	6.9	7.5	7.2	8.1	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.194	0.245	0.271	0.304	0.314	0.311	0.324	0.315	0.288	0.272	0.209	N/A	(40)	
	taud		2.119	2.286	2.295	2.359	2.500	2.541	2.506	2.520	2.580	2.511	2.197	N/A	(41)	
	Ebn at Noon		175	239	271	283	289	291	284	277	268	230	166	N/A	(42)	
	Edh at Noon		10	18	27	31	30	29	29	26	21	15	10	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		83	286	774	1312	1497	1633	1487	1153	744	330	102	44	(44)	
	RadStd		10	52	72	121	130	140	189	104	88	53	23	4	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page